

Coulomb dissociation of high-energy radioactive beams: the case of ^8B

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Coulomb dissociation of ^8B is an alternative method to measure the low-energy $^7\text{B}(p,\gamma)^8\text{B}$ cross section. This reaction is of great current interest in the context of solar neutrinos and has been studied recently both in direct proton-capture [1–4] and in Coulomb-dissociation experiments [5–8]. The validity of the latter method has been questioned since the role of $E2$ multipolarity is considered controversial. The present experiment [8] has studied observables that could clarify the role of a possible $E2$ contribution.

1. THE EXPERIMENT

Coulomb dissociation (CD) of ^8B was performed at GSI at 254 A MeV in a kinematically complete experiment. The ^8B beam was produced by high-energy projectile fragmentation of a ^{12}C beam and separated from the primary beam and contaminant ions with the fragment separator, FRS. After breakup in a 50 mg/cm^2 ^{208}Pb target, the breakup products, ^7Be and p , were detected in two pairs of Si strip detectors and were momentum-analyzed in the kaon spectrometer KaoS [9]. From the momenta and opening angles, the invariant mass of the $^8\text{B}^*$ system prior to breakup could be reconstructed. Using the p and ^7Be momentum vectors, angular distributions have been constructed that are sensitive to the multipolarity of the electromagnetic radiation involved. Thus, the role of $E2$ in the Coulomb breakup of ^8B can be clarified.

To prepare the experiment and to interpret the results, extensive Monte-Carlo simulations have been performed. These simulations are based on a mostly standard CD model by Typel [8]. It includes absorption of $E1$ and $E2$ photons in a first-order perturbation-theory calculation plus an absorptive nuclear potential that takes into account nuclear overlap for small impact parameters. The nuclear structure of ^8B is described by a simple single-particle model using a Woods-Saxon potential. A nonstandard modification of this model is a fit of the potential parameters to the mirror-system, $n+^7\text{Li}$, s -wave scattering lengths [10]. Using this theoretical description as an event generator, the simulation then takes into account the finite detector sizes and all physical processes that affect the momentum and angular resolution, in particular the angular straggling in the Pb breakup target, the Si detectors, and other layers of matter, and the finite position resolution of

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the KaoS detectors.

As a first test of the model, we compare its predictions in the left-hand panel of Fig.1 to the measured θ_8 scattering angle distribution, i.e. the angular distribution of ${}^8\text{B}^*$ with respect to the incident ${}^8\text{B}$ beam. The measured data are very well reproduced if $E1$ multipolarity alone is included in the model. Additional inclusion of $E2$ deteriorates the agreement for $\theta_8 \gtrsim 0.8$ degrees. This observation gives a first hint that $E2$ multipolarity cannot play a major role.

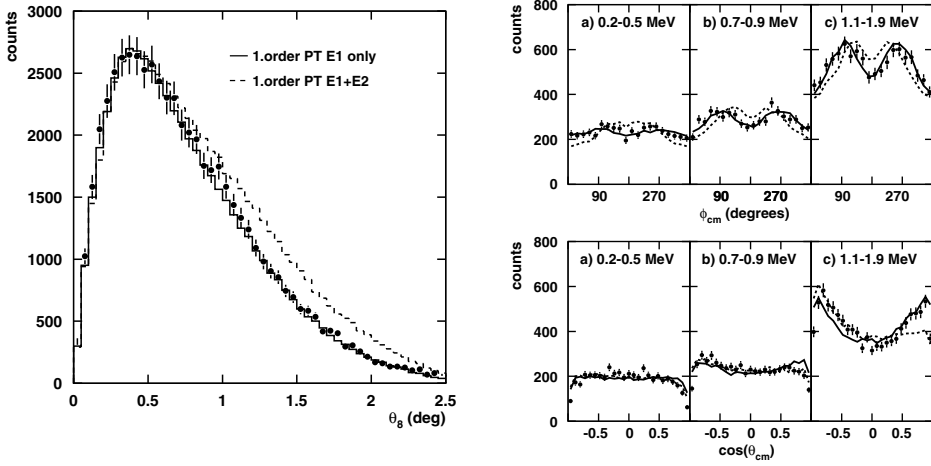


Figure 1. Left: Experimental and simulated θ_8 scattering-angle distributions of the ${}^8\text{B}^*$ nucleus prior to breakup. The full (dashed) histogram has been calculated assuming $E1$ ($E1+E2$) multipolarity in a first-order perturbation-theory calculation. The experimental resolution is included in the histograms. Right: The same for proton azimuthal (top) and polar (bottom) angular distributions for three different bins of p- ${}^7\text{Be}$ relative energy E_{rel} . Full (dashed) lines denote $E1$ ($E1+E2$) multipolarities.

2. CENTER-OF-MASS ANGULAR AND MOMENTUM DISTRIBUTIONS

Sensitive observables to determine the amount of $E1/E2$ mixture are e.g. the center-of-mass angular distributions of the protons after breakup. Our first-order perturbation-theory calculations predict for $E1$ multipolarity symmetric distribution in both the polar and azimuthal angles (right-hand side of Fig.1, full lines). An $E2$ component leads to characteristic asymmetries visualized by the dashed lines. The experimental angular distributions are symmetric, however. This is most clearly seen for the rightmost panels plotted for large p- ${}^7\text{Be}$ relative energies E_{rel} where d -waves contribute most. We conclude that within our experimental errors, the $E2$ contribution can be neglected, and that S_{17}

can be deduced from the energy-differential cross sections under the assumption of pure $E1$ multipolarity, as was earlier done by Kikuchi *et al.* [5] and by Iwasa *et al.* [6].

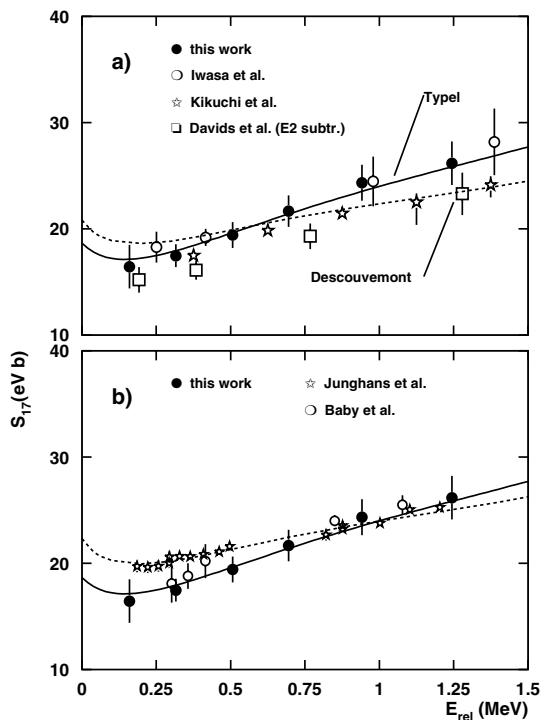


Figure 2. a) S_{17} factors from CD experiments. The filled (open) circles are from the present (previous) GSI experiment. Open stars depict data from Ref.[5], and open squares from Ref.[7]. Theoretical curves by Typel [8] and by Descouvemont [11] have been fitted to the data from this work (full circles). b) Comparison of the present CD results with those of the latest direct-proton-capture experiments (Refs.[3,4]) where the M1-contribution was subtracted. Here the Descouvemont theory has been fitted to the data of Ref.[3].

3. THE ASTROPHYSICAL S_{17} FACTOR

In CD, all impact parameters and $p\text{-}^7\text{Be}$ energies E_{rel} are measured simultaneously, which avoids systematic errors that may influence the results from direct p-capture measurements. The price to be paid, however, is a moderate E_{rel} resolution the magnitude of which can only be estimated from Monte-Carlo simulations of the experiment. We estimate $\sigma_{E_{rel}}$ to be about 40 keV at $E_{rel} = 100$ keV and 140 keV at $E_{rel} = 1500$ keV. To account for this moderate resolution, the astrophysical S -factors $S_{17}(E_{rel})$ are determined by simulating the energy-differential cross sections using our theoretical model and

binning the results with the same bin widths as the experimental cross sections. The experimental S_{17} factors are then determined by multiplication of the theoretical S_{17} factor by the ratio of experimental over theoretical bin contents.

Up to $E_{rel}=1.5$ MeV, the S_{17} factors derived in this way from our data (Fig.2a) agree well with our previous measurements [6] and fit perfectly to the theory of Typel *et al.* [8]. The direct proton capture data, in particular those of Junghans *et al.* [3], are significantly better described by the Descouvemont cluster-model description [11](Fig.2b). Between 0.7 and 1.5 MeV, our data and those of Refs.[3,4] agree well. Below the M1 resonance, however, our data are in clear disagreement with Junghans *et al.*, while they are compatible with the results of Refs.[4] and others (Refs.[1,2]) which are not shown here for clarity.

The different shapes of the theoretical descriptions lead to markedly different extrapolated S factors at zero energy: we find $S_{17}(0) = 18.6 \pm 1.2 \pm 1.0$ eV b extrapolating with the theoretical curve of Typel [8], whereas the most precise direct-capture experiment of Junghans *et al.* obtains $S_{17}(0) = 22.3 \pm 0.7 \pm 0.5$ eV b using the cluster-model theory of Ref. [11]. Only more high-precision measurements with different techniques can resolve these discrepancies. It should be noted that measurements using another indirect approach, the method of Asymptotic Normalization Coefficients (ANC) [12,13] generally gives rather low values of $S_{17}(0)$. E.g. Trache *et al.* [12] find $S_{17}(0) = 17.4 \pm 1.5$ eV b, very similar to the result of Azhari *et al.* of $S_{17}(0) = 17.3 \pm 1.8$ eV b, and fully compatible with our results. The general validity of this method and its accuracy has still to be demonstrated unambiguously, similar to the indirect method of CD used in the present contribution.

REFERENCES

1. F. Hammache *et al.*, Phys. Rev. Lett. **86**, 3985 (2001).
2. F. Strieder *et al.*, Nucl. Phys. **A696**, 219 (2001).
3. A.R. Junghans *et al.*, Phys. Rev. C **68**, 065803 (2003).
4. L.T. Baby *et al.*, Phys. Rev. Lett. **90**, 022501 (2003).
5. T. Kikuchi *et al.*, Phys. Lett. B **391**, 261 (1997);
T. Kikuchi *et al.*, Eur. Phys. J. A **3**, 213 (1998).
6. N. Iwasa *et al.*, Phys. Rev. Lett. **83**, 2910 (1999).
7. B. Davids *et al.*, Phys. Rev. C **63**, 065806 (2001).
8. F. Schümann *et al.*, Phys. Rev. Lett. **90**, 232501 (2003).
9. P. Senger *et al.*, Nucl. Instr. Methods Phys. Res., Sect. B **327**, 393 (1993).
10. L. Koester *et al.*, Z. Phys. A **312**, 81 (1983).
11. P. Descouvemont *et al.*, Nucl. Phys. **A567**, 341 (1994).
12. L. Trache *et al.*, Phys. Rev. Lett. **87**, 271102 (2001).
13. A. Azhari *et al.*, Phys. Rev. C **63**, 055803 (2001).